SOMALA AJAY

AI/ML Engineer | Gen AI Engineer | Python Developer

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PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years of experience building applied GenAI and machine learning systems. Designed a production RAG architecture (FAISS-indexed retrieval + Llama-3.3-70B via Groq) delivering citation-backed QA over a 270-page technical manual, with confidence scoring and multi-turn context handling. Independently architected a full-stack SaaS platform (PinGuru), covering backend, billing, and third-party API integration. Delivered 20+ ML/DL models across NLP and computer vision, including CNN and LSTM-based architectures. Stack: Python, FastAPI, LangChain, FAISS, Docker, SQL.

WORK EXPERIENCE

TCS, *Associate Software Engineer* 12/2025 – 05/2026 | Hyderabad

- Built classification models using Scikit-learn, achieving ~82–87% accuracy on validation datasets
- Designed and implemented end-to-end ML pipelines for structured client datasets — ingestion, preprocessing, and model training Deployed trained ML models via Flask REST API to serve internal inference workflows
- Tracked and versioned ML experiments using Git, ensuring reproducibility across development iterations
- Leveraged GitHub Copilot to accelerate Flask/FastAPI REST API prototyping, cutting boilerplate generation time ~25%
- Performed EDA and feature engineering (Pandas, NumPy) on 10K–50K record datasets, automating preprocessing to cut manual effort ~25–30%

Tech Mahindra, *Associate* 09/2024 – 11/2025 | Hyderabad

- Worked on the client project using the Workflow Management System (WFMS) to handle and process structured data.
- Engineered high-volume unstructured and semi-structured text data pipelines within the Workflow Management System (WFMS) to prepare high-fidelity inputs for training downstream NLP and LLM architectures.
- Performed doc auditing and data correction to maintain high accuracy and data integrity.
- Supported data validation and quality checks, reducing discrepancies and improving workflow reliability.
- Designed and implemented robust data validation algorithms and quality verification workflows, eliminating label noise and reducing feature discrepancies by 20%.

BO IT Solutions Pvt Ltd, *AI/ML Data Scientist* 08/2022 – 06/2024 | Vijaywada

- Delivered 20+ end-to-end AI/ML projects spanning NLP, Computer Vision, Deep Learning, and applied GenAI — from data collection through model deployment — across varied client problem statements
- Built a domain-specific chatbot using GPT-3.5 Turbo, engineering a context-injection pipeline that grounded responses in a structured institutional dataset, with integrated text-to-speech for voice output
- Developed an AI-powered semantic search system using sentence-transformer embeddings and vector similarity matching for meaning-based query retrieval over structured data
- Built a voice-driven AI assistant combining speech recognition and TTS for hands-free interaction
- Specialized in deep learning architectures: built CNN models for image classification and LSTM models for sequence prediction, outperforming traditional ML baselines by ~15%
- Explored both traditional ML (Scikit-learn) and deep learning (TensorFlow) approaches per project, selecting the right technique for each problem — achieving 80–90% accuracy across deliveries Handled large-scale data exploration (10K–100K+ records) with Pandas, NumPy, and Matplotlib, cutting preprocessing time by ~30% across the project portfolio

EDUCATION

B.Tech, *Aditya University* 2018 – 2022 | Surampalem, Andhra Pradesh
Electronics And Communication Engineering

TECHNICAL SKILLS

Generative AI & Agents — LLMs, RAG Pipelines, Semantic Seacrh, Vector Databases, LangChain, LlamaIndex, Prompt Engineering, Agentic Workflows, ADK & MCP

Programming & Backend — Python, SQL, FastAPI, Flask, REST APIs

Machine Learning & DL — Scikit-learn, TensorFlow, Keras, CNN, LSTM, NLP, Feature Engineering, Model Evaluation

Data Engineering & DBs — Data Validation, Data Preprocessing, ETL Pipelines, SQL, Pandas, NumPy, MongoDB, MySQL,

Cloud & Developer Tools — Docker, Git, VS Code, GitHub Copilot, GCP, AWS, DigitalOcean

HACKATHON PARTICIPATION

National AI Hackathon - i2e Consulting Bangalore
Project: Complex Technical Manual QA System using RAG

- Built a production-grade RAG system ingesting and indexing 885 semantic vectors from the 270-page NASA Systems Engineering Handbook using FAISS and sentence-transformers (all-MiniLM-L6-v2)
- Integrated Groq API (LLaMA-3.3-70B) with confidence scoring (0–100%), multi-turn conversation memory, section-level citations, and NASA acronym expansion (20+ terms: TRL, PDR, KDP, SRR)

Stack: FastAPI | FAISS Groq API | LLAMA 3.3-70B | sentence-transformers | PyPDF | Vanilla JS | Python | PdfPLumber

KEY PROJECTS

PinGuru — Instagram DM Automation SaaS,
Python | FastAPI | MongoDB | Instagram Graph API | DigitalOcean | Stripe | GitHub Pages

- Architected and deployed **Pinguru**, a full-stack Instagram automation SaaS (pinguru.me) featuring real-time DM triggers and multi-tier Stripe billing. Built with a **FastAPI/MongoDB** backend and **Instagram Graph API**, the platform integrates robust security (JWT, OTP) and automated comment-to-DM workflows.

Complex Technical Manual QA System Using RAG, *i2e-Consulting*
https://github.com/Ajaysomala/i2e_Hireathon-Complex_Technical_Manual_QA_System

- NASA Manual QA System | RAG Pipeline, Python | FastAPI | FAISS | Llama 3.3 | Groq | NLP. Designed a production-grade RAG pipeline using **FastAPI**, **FAISS**, and **Llama 3.3** to automate citation-backed QA for the 270-page NASA Systems Engineering Handbook.

Image Caption Generator, *Python | CNN | LSTM | Deep Learning | NLP | FLICKR DataSet*

- Built an end-to-end image captioning system using a CNN encoder for visual feature extraction and an LSTM decoder for sequential caption generation, trained on the Flickr8k dataset

Autonomous Multi-Agent Operations System,
Python | FastAPI | Groq API (Llama 3.3) | Multi-Agent Orchestration | Docker | SQLite | Pydantic

- Built a multi-layered AI agent system with increasing autonomy. A reflex layer handles known triggers with fixed rules. A single AI agent (Llama 3.3) interprets plain-language instructions and plans its own steps. For complex goals, a team of agents — Planner, Executor, and Reviewer — split the work, execute it, and self-correct through review. The whole system runs as a containerized FastAPI service with task memory.

CERTIFICATIONS

Generative AI Introduction
Microsoft Learning

Career Essentials in Generative AI by Microsoft and LinkedIn
Microsoft

Develop an AI app with the Azure AI Foundry SDK
Microsoft

Learning Microsoft 365 Copilot for Work
Linkedin